

Photonic Devices - Gyroscope report

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I. INTRODUCTION

In this report, we comment on the laboratory experience dealing with the fiber-optic gyroscope (FOG), a sensor used for measuring changes in orientation. Applications of such sensor include inertial navigation systems, such as in the Hubble telescope or airplane flight. The performance of a gyroscope mainly depends on two factors:

- Scale factor: the coefficient that converts the measured quantity (response) into rotation rate.
- Rotation-rate accuracy: very small rotations require high accuracy.

A. Sagnac Effect

To illustrate the principle behind this sensor, a circular loop is considered: the time of flight through the loop at rest is $T_0 = 2\pi R/c$.

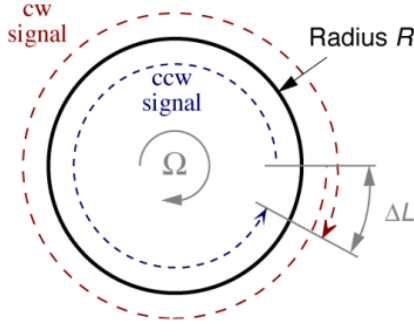


Fig. 1: Sagnac Loop.

If the loop is rotating clockwise around its axis at rate Ω , the CW signal has to cover a distance of $2\pi R$ plus the distance by which the loop rotated in the time T . Similarly, the CCW signal takes a shorter time to complete the loop.

As a consequence of the loop rotation, the light beams propagating in opposite directions in a rotating frame experience a difference in optical path length and a difference in phase shift, proportional to the rotation rate and to the loop area. The phase shift is given by

$$\phi_s = \frac{8\pi\Omega A_{loop}}{c\lambda}. \quad (1)$$

If the medium through which light is propagating is different from vacuum (i.e. refractive index $n \neq 1$), two new effects come into play: the beams travel at a lower velocity (c/n) and the medium drags the two waves (Fresnel-Fizeau drag). Overall, these two effects cancel each other out, hence ϕ_s is independent from the refractive index of the medium.

B. Sagnac Interferometer

By exploiting the Sagnac effect, we were able to develop a lossless interferometer from which we measured the output power in order to evaluate the phase shift and thus the corresponding rotation, as given by (1).

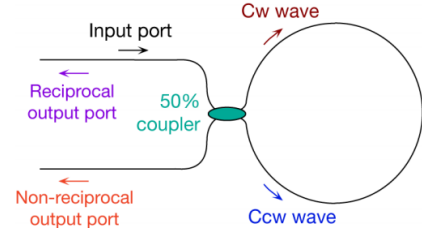


Fig. 2: Sagnac interferometer.

In Fig. 2, the coupler has a 50/50 power output in a 2x2 configuration. Since the device is lossless, the power reflected at the reciprocal port is $|b_1|^2 = 0.25 \cos^2(\phi_s/2)|a_1|^2$. From this equation we can plot the relationship between the Sagnac phase shift and the output power from the interferometer:

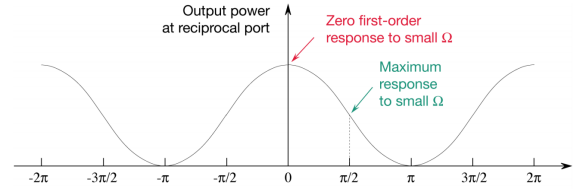


Fig. 3: Sagnac Phase Shift.

The FOG basic response is nonlinear and periodic, and the sensitivity to the rotation rate is zero for very small angular variations. In order to achieve a higher sensitivity, the interferometer must be biased at $\pi/2$, where the sensitivity is at the highest point; this can be done by means of a phase modulator.

II. EXPERIMENTAL SETUP

In the typical setup seen during the lectures (shown in Figure 4) a coherent, unpolarized laser source is connected to a 50/50 coupler, where the other input port is connected to the photo-detector that measures the response of the sensor. In this experience, the setup is slightly different, in that the first coupler is replaced by a circulator, so that the light coming from the Erbium-doped fiber amplifier (EDFA) goes towards the 50/50 loop coupler, while the backward-propagating exits from another port. The setup also features a polarization controller, inserted before the loop coupler, which polarizes light to enable interference. Fortunately, the fiber loop is only slightly affected by PMD, hence polarization remains

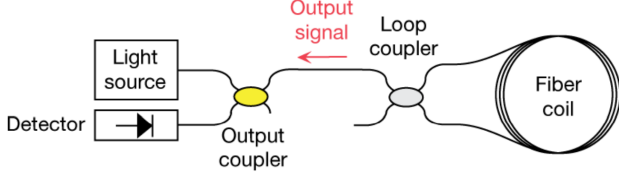


Fig. 4: Experimental setup seen during the lectures. The actual setup features an EDFA, a circulator and a polarization controller; moreover, the output is measured at the nonreciprocal port.

relatively stable throughout the bobbin. After propagation in the loop, the interfering clockwise and counterclockwise signals are measured at the non-reciprocal output port for convenience, although the reciprocal port should be used to avoid fading of the output signal; hence, the reciprocal output port was not used.

The fiber coil was placed over a motorized platform which was controlled through a computer interface on Simulink. We were thus able to choose the angular velocity, the direction and the duration of the rotation to impress on the coil. Moreover, we were able to modify motion to have constant angular velocity (for the first part of the experience) and constant angular acceleration (for the second part).

The detector has different levels of sensitivity. For a given rotational speed of the coil, we have to calibrate the sensor range in order to read the correct output power.

III. RESULTS

A. Constant angular velocity

In the first part of this experience, the given spreadsheet was filled with data. The fiber coil was spun at 35 different rotation rates between -15 and +15 rpm (revolutions per minute). The rotation time was adjusted according to the velocity, in order not to make the bobbin spin too fast and risk of entangling it, but it was between 10 and 20 seconds for all the measures.

For each rotation rate, a MATLAB file was generated from the oscilloscope data. Each file contains information about the rotation angle ϕ , the rotation rate Ω (both the empirical and the nominal value), the voltage applied to the motor and, most importantly for our analysis, the readout voltage V measured during the rotation. The obtained data of channels 1, 2 and 4 with a rotation rate of $\Omega = 10$ rpm is shown in figure 5 as an example.

To fill the spreadsheet, we first obtained the time-average readout voltage V for each rotation rate, together with the range of the power meter r . The output power was then computed by

$$P_{out} = \frac{V \cdot 10^{-r}}{R}, \quad (2)$$

where R is the responsivity of the detector, whose value is shown in Table I, together with other relevant parameters of this experiment.

The resulting graph of the output power as a function of the rotation speed is shown in Figure 6. Since, in the case of this

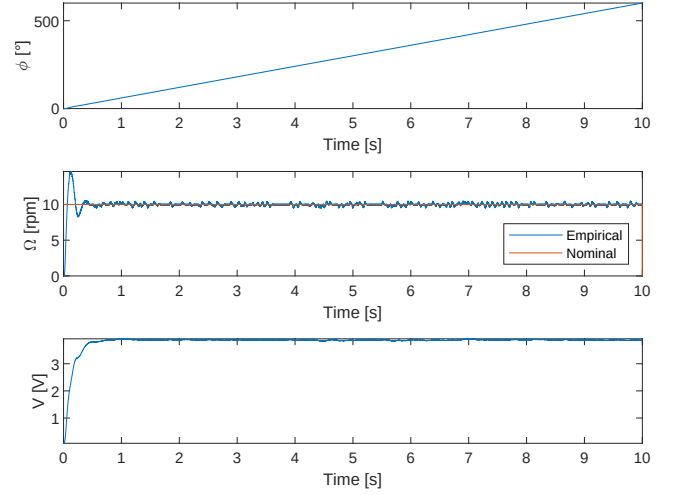


Fig. 5: Example of output data for $\Omega = 10$ rpm.

Parameter	Symbol	Value
Fiber length	L	1160 [m]
Bobbin radius	R_B	81 [mm]
Optical wavelength	λ	1550 [nm]
Responsivity	R	0.01 [mA/mW]
Number of turns	N	2279

TABLE I: Setup parameters.

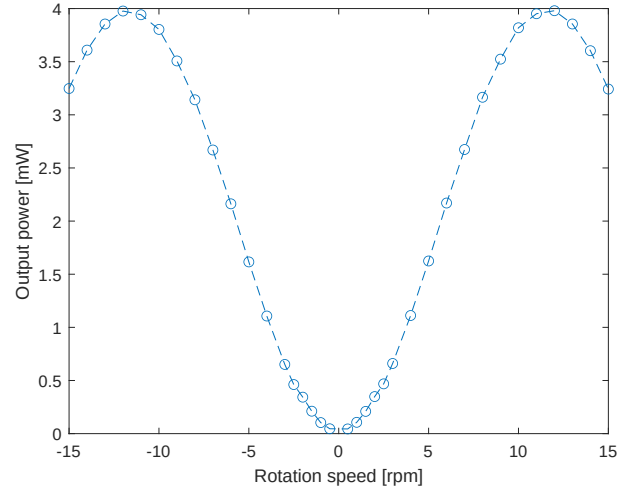


Fig. 6: Output power versus rotation speed.

experiment, the power values are referred to the nonreciprocal port, we obtain a squared-sinusoidal function, instead of a squared-cosinusoidal as in a conventional FOG. Nonetheless, the characteristic shape is quite clear, and the results confirm the expected behavior of this system. The output power is represented by a raised sinusoidal function, with an expected minimum of 0 (corresponding to the state in which the Sagnac interferometer is at rest) and a maximum of about 4 mW. Comparing this figure with the theoretical one (Figure 3) exploit the fact that we are using the non-reciprocal port and so on the phase shift is translated by a factor π .

Finally, as requested, we evaluated both the empirical and theoretical value of the so-called gyroscope scale factor. The

former was inferred via Figure 6, by observing that half a period (corresponding to a shift of π radians) is obtained at 12 rpm ≈ 1.256 rad/s. The latter was simply calculated via (1) and the parameters in Table I. The result gives

$$\frac{\phi_s}{\Omega} = \frac{\pi}{1.256 \text{ s}^{-1}} = 2.50 \text{ s} \approx 2.54 \text{ s} = \frac{8\pi N A_{loop}}{c\lambda}. \quad (3)$$

This factor gives the average linear sensitivity of the phase shift with respect to the rotation rate of the gyroscope.

B. Constant angular acceleration

In the second part of the experience, the Simulink setup was changed to provide a constant angular acceleration of $\alpha = 1.5$ rpm/s ≈ 0.157 rad/s² to the fiber reel. The output data from such configuration is shown in Figure 7.

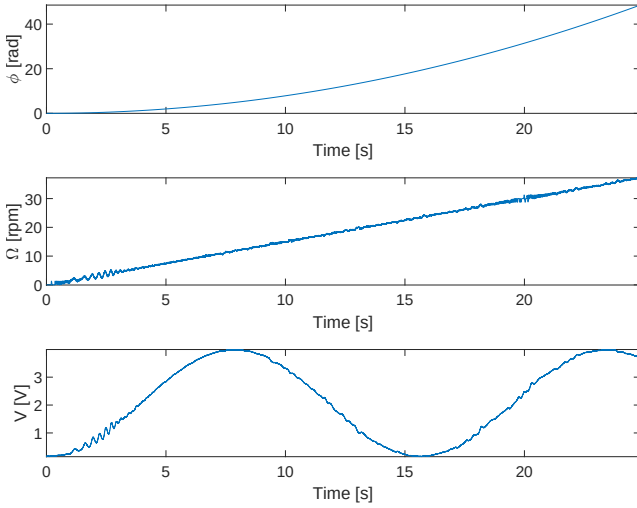


Fig. 7: Output data with constant acceleration.

This approach is particularly useful because it directly shows the relation between the behavior of the spinning rate and the output voltage (and therefore the power). Naturally the former grows linearly, as shown in the figure, because of the constant acceleration. The latter, on the other hand, gives a similar sinusoidal pattern to that manually obtained in the first part of the experience.

The last part of the data analysis was thus focused on proving this intuitive relation between the rotation rate and the output voltage. To do so, the voltage signal $V(t)$ was normalized to be truly cosinusoidal by using this relation:

$$V'(t) = 1 - \frac{V(t)}{2\max(V)}. \quad (4)$$

Having processed the voltage, the relation between the cosine of the spinning rate $\cos(a\Omega(t))$ (where a is a scale factor) versus the voltage $V'(t)$ was plotted on a graph, which is reported in Figure 8.

IV. CONCLUSION

In Section I, we introduced the definition and applications of a gyroscope. Then, we reported the working principle of a Sagnac interferometer which is able to detect the amplitude

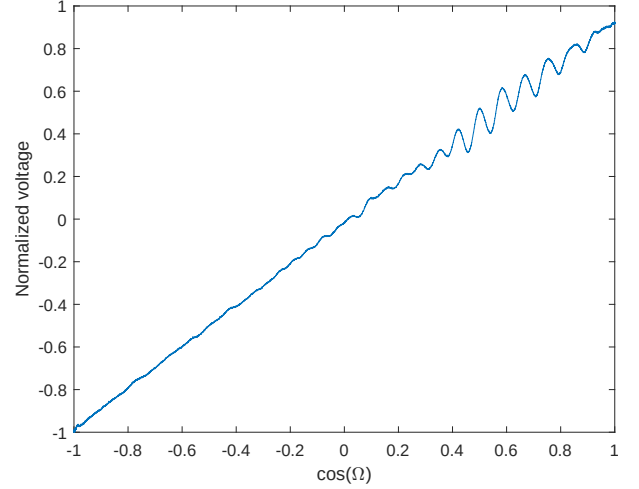


Fig. 8: Linear relation between Ω and V' .

of the rotation through (1). The principle at the basis of the sensor is the following: if the same optical signal is injected in a fiber loop both in clockwise and counterclockwise direction (by using a coupler), then due to the loop rotation the output phase will be different for the two signals.

Then, in Section II we developed a system able to detect the accuracy and performance of our optical gyroscope thanks to a motorized platform and a power detector. We collected data for different constant rotation rates and for an accelerating rotation rate scenario.

In the results (Sec. III), we discovered that the gyroscope factor, given in (3), of our setup is about 2.50 s, and thus matches the theoretical one, estimated at 2.54 s.

Furthermore, by comparing Figure 3 and Figure 6, we achieved the expected relation between power and phase shift (our graph is translated due to the use of the non-reciprocal port instead of the reciprocal one).

We also exploited the correlation between the parameters to show that the cosine of the rotation angle and the output voltage are linearly related (Fig. 8) in the accelerating setup.

Compared to what we said in the introduction, these performances are suitable for a inertial navigation grade gyroscope.

However, this simple FOG is capable only to detect rotation around the axis orthogonal to its rest plane: rotation around the other two axes, are not sensed.

Another weak point of the FOG is that the measure is sensitive to the transient effects of the Sagnac interferometer. As a matter of fact, if there is a transient phase perturbation applied asymmetrically in the loop (i.e. not at the loop's midpoint) it will be seen by the CW and CCW waves at different times. This causes a transient power change at the output ports. If this perturbation varies rapidly (e.g. because of vibrations), noise is induced in the output, while if it varies slowly (e.g. because of a change in temperature), it will induce a drift. Nonetheless, these effects did not compromise the reliability of the measurements, as the plot of the power against the rotation speed and the other graphs obtained from the data have shown.